Appendices

Appendix A

Some results from Calculus on $\mathbb{R}^n$

A.1 Results from Differentiation

Theorem A.1.1. ***(Taylor's Theorem)*** *Let $f : U \to \mathbb{R}$ be C^1 in a neighbourhood U of $p \in \mathbb{R}^n$ that is star shaped. Then there exists $g_1, g_2, \cdots g_n$ such that*

$$f(x) = f(p) + \sum_{i=1}^{n} g_i(x)(x^i - p^i)$$

where $g_i(p) = \frac{\partial}{\partial x^i}|_p f$

Proof. Let $x \in U$. Since U is star shaped, we have $\gamma(t) = p + t(x - p) \in U$ for $t \in [0, 1]$. Consider $\frac{d}{dt}(f \circ \gamma(t)) = Df(\gamma(t)) \cdot \gamma'(t) = \sum_{i=1}^{n} \frac{\partial}{\partial x^i} f(\gamma(t))(x^i - p^i)$. Integrate both sides to get

$$\int_{t=0}^{t=1} \frac{d}{dt}(f \circ \gamma(t))dt = f(x) - f(p) = \sum_{i=1}^{n}(x^i - p^i) \int_{t=0}^{t=1} \frac{\partial f}{\partial x^i}(p + t(x - p))dt$$

And let $g_i(x) := \int_{t=0}^{t=1} \frac{\partial f}{\partial x^i}(p + t(x - p))dt$ whence it is easy to verify that $g_i(p) = \frac{\partial f}{\partial x^i}(p)$.

■

Theorem A.1.2. ***(Inverse Function Theorem)*** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. Suppose that $\det Df(a) \neq 0$ for a given $a \in \mathbb{R}^n$. Then, there exists U and V open in $\mathbb{R}^n$ such that $a \in U$, $f(a) \in V$ and $f|_U : U \to V$ is a diffeomorphism.*

Theorem A.1.3. ***(Implicit Function Theorem)*** *Let $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ be continuously differentiable in an open set of (a, b) and let $f(a, b) = 0$. Let M be the $m \times m$ matrix*

$$D_{n+j} f^i(a, b); 1 \leq i, j \leq m$$

and suppose $\det M \neq 0$. Then there exists open sets $A \subset \mathbb{R}^n$, $B \subset \mathbb{R}^n$ with $a \in A, b \in B$ such that for all $x \in A$ there is a unique $g(x) \in B$ so that $f(x, g(x)) = 0$, and the g so obtained is differentiable.

Proof. Define $F : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n \times \mathbb{R}^m$ given by $F(x, y) := (x, f(x, y))$. This function is clearly differentiable, and moreover,

$$DF = \begin{pmatrix}
\frac{\partial x^1}{\partial x^1} & \frac{\partial x^1}{\partial x^2} & \cdots & \frac{\partial x^1}{\partial x^n} & \frac{\partial x^1}{\partial y^1} & \frac{\partial x^1}{\partial y^2} & \cdots & \frac{\partial x^1}{\partial y^m} \\
\frac{\partial x^2}{\partial x^1} & \frac{\partial x^2}{\partial x^2} & \cdots & \frac{\partial x^2}{\partial x^n} & \frac{\partial x^2}{\partial y^1} & \frac{\partial x^2}{\partial y^2} & \cdots & \frac{\partial x^2}{\partial y^m} \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\
\frac{\partial x^n}{\partial x^1} & \frac{\partial x^n}{\partial x^2} & \cdots & \frac{\partial x^n}{\partial x^n} & \frac{\partial x^n}{\partial y^1} & \frac{\partial x^n}{\partial y^2} & \cdots & \frac{\partial x^n}{\partial y^m} \\
\frac{\partial f^1}{\partial x^1} & \frac{\partial f^1}{\partial x^2} & \cdots & \frac{\partial f^1}{\partial x^n} & \frac{\partial f^1}{\partial y^1} & \frac{\partial f^1}{\partial y^2} & \cdots & \frac{\partial f^1}{\partial y^m} \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\
\frac{\partial f^m}{\partial x^1} & \frac{\partial f^m}{\partial x^2} & \cdots & \frac{\partial f^m}{\partial x^n} & \frac{\partial f^m}{\partial y^1} & \frac{\partial f^m}{\partial y^2} & \cdots & \frac{\partial f^m}{\partial y^m}
\end{pmatrix}$$

And it is easy to see that $\det DF(a,b) = detM \neq 0$. Therefore, from Inverse function theorem (with litte modification), there exists an open set $A \times B$ of $\mathbb{R}^n \times \mathbb{R}^m$, and an open set $W \subset \mathbb{R}^n \times \mathbb{R}^m$ such that $(a,b) \in A \times B$, $(a,0) \in W$ and $F : A \times B \to W$ is a diffeomorphism. $F^{-1} : W \to A \times B$ is also a diffeomorphism. Moreover, note that $F^{-1}(p,q)$ is of the form $(p, k(p,q))$ for some differentiable function $k : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$. $\pi \circ F(x,y) = \pi \circ (x, f(x,y)) = f(x,y)$ so $\pi \circ F \equiv f$. Look at $f \circ (p, k(p,q)) = f \circ F^{-1}(p,q) = \pi(p,q) = q$ so $f \circ (p, k(p,0)) = 0$. Our $g(p)$ is given by $k(p,0)$, and it is differentiable as a composition of differentiable functions.

■

Theorem A.1.4. ***(A corollary of A.1.3)*** *Let $f : \mathbb{R}^n \to \mathbb{R}^p (p \leq n)$ be continuously differentiable in an open set containing $a \in \mathbb{R}^n$. If $f(a) = 0$ and the matrix $D_j(f^i(a))_{p \times n}$ has rank p, then there exists an open neighbourhood A of a and a diffeomorphism $h : A \to h(A) \subset \mathbb{R}^n$ such that $f \circ h(x_1, x_2 \cdots x_n) = (x_{n-p+1}, x_{n-p+2} \cdots, x_n)$.*

Proof. Consider $f : \mathbb{R}^{n-p} \times \mathbb{R}^p \to \mathbb{R}^p$. Suppose that the matrix $Q = D_{n-p+i}(f^j); i,j \leq p$ be invertible ($\det Q \neq 0$). Let, as before, $F : \mathbb{R}^{n-p} \times \mathbb{R}^p \to \mathbb{R}^{n-p} \times \mathbb{R}^p$ given by

$$F(p,q) = (p, f(p,q))$$

Since $\det DF \neq 0$, there exists, around $a = (a^1, a^2, \cdots, a^{n-p}, a^{n-p+1}, \cdots a^n)$ a neighbourhood $A_{n-p} \times A_p \subset \mathbb{R}^{n-p} \times \mathbb{R}^p$ and a neighbourhood $W \subset \mathbb{R}^{n-p} \times \mathbb{R}^p$ containing $F(a) = (a_1, a_2 \cdots a_{n-p}, 0)$ such that $F : A_{n-p} \times A_p \to W$ is invertible and whose inverse is continuously differentiable. Note that, the inverse of F ought to be of the form $F^{-1}(x_1, x_2, \cdots, x_{n-p}, y_{n-p+1}, \cdots, y_n) = (\vec{x}, b(\vec{x}, \vec{y}))$ (let $\vec{x} = (x_1, x_2, \cdots x_{n-p}) \in \mathbb{R}^{n-p}$ and $\vec{y} = (y_{n-p+1}, y_{n-p+2}, \cdots, y_n) \in \mathbb{R}^p$) where it is understood that $b : \mathbb{R}^n \to \mathbb{R}^p$ is continuously differentiable. Also note that $\pi_2 \circ F \equiv f$. We have that

$$f \circ F^{-1}(x_1, x_2 \cdots x_{n-p}, x_{n-p+1}, \cdots x_n) =$$

$$\pi_2(x_1, x_2, \cdots x_{n-p}, x_{n-p+1}, \cdots x_n) = (x_{n-p+1}, x_{n-p+2}, \cdots, x_n)$$

Case 2: If the indices where the matrix $\{(\partial_i f^j(a)); 1 \leq j \leq p, i = i_1, i_2, \cdots i_p\}$ is not the last p indices, we write the linear transformation $\text{Swap} : \mathbb{R}^n \to \mathbb{R}^n$ that takes these p indices to the last p indices. This is a linear transformation that is invertible since it takes basis elements to basis elements. If we look then, at $f \circ \text{Swap} : \mathbb{R}^n \to \mathbb{R}^p$, we see that this is a function of the form of Case 1, so we have a h such that $f \circ \text{Swap} \circ h(x_1, x_2, \cdots x_n) = (x_{n-p+1}, x_{n-p+2}, \cdots x_n)$ which means our desired function is $\text{Swap} \circ h$.

■

A.2 Results from Integration

A.2.1 The Riemann Integral

Definition A.2.1. (Partition) For closed interval $[a,b]$ in $\mathbb{R}$, a partition is simply a collection of sub-intervals $[a_i, a_{i+1}]$ so that $a_0 = a < a_1 < a_2 < \cdots a_k = b$.

For a closed rectangle R in $\mathbb{R}^n$, a partition P, usually written along with its subpartitions (as in, $P \equiv (P_1, P_2, \cdots, P_n)$), is *a* collection of subrectangles R_i (of a particular form) so that $R = \cup_i R_i$ and $R_i^\circ \cap R_j^\circ = \emptyset$. Explicitly, if

$$R = [a^1, b^1] \times [a^2, b^2] \times \cdots \times [a^n, b^n]$$

,

$$P_i \equiv [a^i = a^i_0 < a^i_1 < a^i_2 \cdots a^i_{k_i} = b^i]$$

and I^i_j denotes the $j-$th subinterval of the $i-$th partition, the collection of all possible sub rectangles of R of the form $\{I^1_{j_1} \times I^2_{j_2} \times \cdots \times I^n_{j_n} : 1 \leq j_q \leq \text{number of intervals}(P_q)\}$ is a partition of R.

Definition A.2.2. (Upper and Lower sums) Let R be a rectangle in $\mathbb{R}^n$, $f : R \to \mathbb{R}$ be a bounded function and let P be a partition of R.

$$\mathcal{L}(P, f) := \sum_{S \in P (S \text{ is a subrectangle})} (\inf_{x \in S} f(x))|S|$$

$$\mathcal{U}(P, f) := \sum_{S \in P (S \text{ is a subrectangle})} (\sup_{x \in S} f(x))|S|$$

where $|S|$ is the volume of the subrectangle S. For convenience, we denote $\inf_{x\in S} f(x)$ by $m_S(f)$ and $\sup_{x\in S} f(x) = M_S(f)$.

Lemma A.2.3. *For a bounded function $f : R \to \mathbb{R}$,*

$$\mathcal{L}(P, f) \leq \mathcal{U}(P, f)$$

for a given partition P.

> *Proof.* If S_i are subrectangles of S coming from P', then $m_S(f)|S| = m_S(f)(|S_1|+|S_2|+\cdots+|S_i|) \leq \sum_{S_i \in S} m_{S_i}(f)|S_i|$. This proves the result for lower sums, and the proof for upper sums is similar.

■

Lemma A.2.4. *If P' is a refinement of P (i.e. each subrectangle of P' is contained in some subrectangle of P) then*

$$\mathcal{L}(P, f) \leq \mathcal{L}(P', f)$$

and

$$\mathcal{U}(P', f) \leq \mathcal{U}(P, f)$$

Lemma A.2.5. *If P and P' are any two partitions and $f : R \to \mathbb{R}$ a bounded function, then $\mathcal{L}(P', f) \leq \mathcal{U}(P, f)$.*

> *Proof.* Let P'' be a partition of both P and P' (i.e. the subrectangles of P'' are contained in some subrectangle of P as well as some subrectangle of P'). Then by the previous lemma,
>
> $$\mathcal{L}(P', f) \leq \mathcal{L}(P'', f) \leq \mathcal{U}(P'', f) \leq \mathcal{U}(P, f)$$

■

Now we are in a position to define the Riemann Integral.

Definition A.2.6. ($\int_R f$) Let $f : R \to \mathbb{R}$ be a bounded function from a rectangle R. We say f is *Riemann integrable* if

$$\sup_P \mathcal{L}(P, f) = \inf_P \mathcal{U}(P, f)$$

where sup and inf are taken over all partitions P of R. In that case we say $\sup_P \mathcal{L}(P, f) = \inf_P \mathcal{U}(P, f) := \int_R f$.

Lemma A.2.7. *A bounded function $f : R \to \mathbb{R}^n$ is Riemann Integrable if and only if for every $\varepsilon > 0$, there exists a partition P such that $\mathcal{U}(P, f) - \mathcal{L}(P, f) < \varepsilon$.*

A.2.2 Measure Zero and Content Zero

Definition A.2.8. (Measure Zero) A set $A \subset \mathbb{R}^n$ is said to be *measure zero* if for every $\varepsilon > 0$, there exists a countable collection of closed rectangles R_i such that $A \subset \bigcup_i R_i$ and $\sum_i |R_i| < \varepsilon$.

Lemma A.2.9.

1. *Subsets of Measure Zero sets are Measure Zero.*
2. *Countable union of Measure Zero sets is Measure Zero.*

Proof. (1) Is trivial.

For (2), consider E_i a countable collection of measure zero sets. Let $\varepsilon > 0$. Consider $\varepsilon/2^i$. There exists a collection of closed rectangles $\{R_{ij} : j \in \mathbb{N}\}$ such that $E_i \subseteq \bigcup_j R_{ij}$ and $\sum_j |R_{ij|} < \varepsilon/2^i$. Look at the collection $\{R_{ij} : i, j \leq \mathbb{N}\}$ as a whole. This is a countable collection of closed rectangles that cover $\cup_i E_i$, and note that $\sum_{i,j} |R_{i,j}| < \varepsilon$.

∎

Definition A.2.10. (Content Zero) A set $A \subset \mathbb{R}^n$ is said to have *content zero* if for every $\varepsilon > 0$, there exists a *finite* collection of closed rectangles R_i such that $A \subseteq \cup_{i=1}^n R_i$ and $\sum_{i=1}^n |R_i| < \varepsilon$.

Lemma A.2.11. *If A is compact and has measure zero, then it has content zero.*

A.2.3 Integrable Functions

Definition A.2.12. (Oscillation) Let $f : X \to \mathbb{R}$ be a bounded function, where X is a metric space. Define

$$M(f, a, \delta) := \sup_{x \in B_\delta(a)} f(x)$$

and

$$m(f, a, \delta) := \inf_{x \in B_\delta(a)} f(x)$$

Let

$$\operatorname{osc}(f, a, \delta) := M(f, a, \delta) - m(f, a, \delta)$$

Note first, that $\operatorname{osc}(f, a, \delta)$ is decreasing for decreasing δ. Now we define

$$\boxed{\operatorname{osc}(f, a) := \lim_{\delta \to 0} \operatorname{osc}(f, a, \delta)}$$

An alternative characterisation of oscillation is the following:

Lemma A.2.13. *Let f be as in definition A.2.12. The quantity $\sup_{x,y \in B_\delta(a)} |f(y) - f(x)|$ is decreasing for decreasing δ and*

$$\operatorname{osc}(f, a) = \lim_{\delta \to 0} \Big(\sup_{x,y \in B_\delta(a)} |f(x) - f(y)| \Big) = \inf_{\delta > 0} \Big(\sup_{x,y \in B_\delta(a)} |f(x) - f(y)| \Big)$$

Lemma A.2.14. *$f : X \to \mathbb{R}$ (where X is a metric space) is continuous at a if and only if $\operatorname{osc}(f, a) = 0$*

Proof. $\implies$) Let f be continuous at a. For any given $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in B_\delta(a)$ we have

$$|f(x) - f(a)| < \varepsilon/2 \text{ and } |f(y) - f(a)| < \varepsilon/2$$

Adding these we get

$$|f(x) - f(y)| < \varepsilon$$

for $x, y \in B_{\delta(\varepsilon)}(a)$. Therefore, $\sup_{x,y \in B_{\delta(\varepsilon)}(a)} |f(x) - f(y)| < \varepsilon$. Since for every $\varepsilon > 0$, there exists some $\delta(\varepsilon) > 0$ such that $\sup_{x,y \in B_{\delta(\varepsilon)}(a)} |f(x) - f(y)| < \varepsilon$, we have $\operatorname{osc}(f, a) = 0$.

$\Longleftarrow$) Let $\operatorname{osc}(f, a) = \inf_{\delta > 0}(\sup_{x,y \in B_\delta(a)} |f(x) - f(y)|) = 0$. This means that for every $\varepsilon > 0$, there exists $\delta > 0$ such that $\sup_{x,y \in B_\delta(a)}(|f(x) - f(y)|) < \varepsilon$, which means that

$$|f(x) - f(a)| < \varepsilon$$

whenever $x \in B_\delta(a)$.

■

Lemma A.2.15. (Recharacterisation of oscillation in terms of rectangles in $\mathbb{R}^\mathbf{n}$)

$$\operatorname{osc}(f, a) = \inf_{\delta > 0}\left(\sup_{x \in R_\delta(a)} f(x) - \inf_{x \in R_\delta(a)} f(x)\right)$$

Where $R_\delta(a)$ is a cube centreed at a with side length δ.

Proof. Note that

$$B_{\delta/\sqrt{n}(a)} \subseteq R_\delta(a) \subseteq B_{\delta\sqrt{n}}(a)$$

From here the proof is trivial.

■

Lemma A.2.16. *Let A be a closed rectangle and $f : A \to \mathbb{R}$ be a bounded function such that* $\operatorname{osc}(f, a) < \varepsilon$ *for every $a \in A$. Then there exists a partition P of A such that* $\mathcal{U}(f, P) - \mathcal{L}(f, P) < \varepsilon \cdot |A|$

Proof. We are given a fixed $\varepsilon_0 > 0$. We have that $\operatorname{osc}(f, a) < \varepsilon_0$ for all $a \in A$. $\inf_{\delta > 0}(\sup_{x \in R_\delta(a)} f(x) - \inf_{y \in R_\delta(a)} f(y)) < \varepsilon_0$ for every $a \in A$. This means that, for every $a \in A$, there exists $\delta_a > 0$ such that $\sup_{x \in R_\delta(a)} f(x) - \inf_{x \in R_\delta(a)} f(x) < \varepsilon_0$. Collect all such $R_{\delta_a}(a)$. This is a cover of A, for which a finite subcover $\{R_{\delta_{a_j}}(a_j) : j \leq N\}$ exists. Consider the partition P of A formed by extending the grids of all the cubes $\{R_{\delta_{a_j}}(a_j) : j \leq N\}$. The final partition P's subrectangles J_i have the property that

$$\sup_{x \in J_i} f(x) - \inf_{x \in J_i} f(x) < \varepsilon_0$$

Essentially,

$$M_S(f) - m_S(f) < \varepsilon_0$$

for S (a subrectangle) $\in P$. This gives us that $\mathcal{U}(f, P) - \mathcal{L}(f, P) = \sum_{S \in P}(M_S(f) - m_S(f))|S| < \varepsilon_0 \sum_{S \in P} |S| = \varepsilon_0|A|$

■

Theorem A.2.17. *Let A be a closed rectangle and $f : A \to \mathbb{R}$ be a bounded function. Let $B := \{x \in A : f$ not continuous at $x\}$. Then f is integrable if and only if B is a set of measure* 0.

Proof. Note that $B := \{x \in A : \operatorname{osc}(f, x) > 0\}$ which could be written as $\cup_{n \in \mathbb{N}}\{x \in A : \operatorname{osc}(f, x) \geq 1/n\}$, where we note that $\{x \in A : \operatorname{osc}(f, x) \geq t\}$ is closed for any $t > 0$ (If $\operatorname{osc}(f, a) < \varepsilon$ then there would exist a ball around a such that $osc(f, x) < \varepsilon$ for x in said ball).
$\Longrightarrow$) Let B be measure 0. Consider the closed (and hence compact) set $B_\varepsilon := \{x \in A : \operatorname{osc}(f, x) \geq \varepsilon\}$ whose measure (since $B_\varepsilon \subseteq B$) is 0. We thus have a finite collection of rectangles R_i that cover B_ε and $\sum_{i=1}^N |R_i| < \varepsilon$. Consider the partition P of A generated by extending the grid formed

by R_i and collecting all subrectangles K_i. These can be classified into two subcategories (which together partition A):

1. $(\mathbb{S})^1 := \{K_i : K_i \cap B_\varepsilon \neq \varnothing\}$
2. $(\mathbb{S})^2 := \{K_i : K_i \cap B_\varepsilon = \varnothing\}$

We now look at $\mathcal{U}(f,P) - \mathcal{L}(f,P) = \sum_{S\in(\mathbb{S})^1\cup(\mathbb{S})^2}(M_S(f) - m_S(f))|S|$.

Look at the first sum:

$$\sum_{S\in(\mathbb{S})^1}(M_S(f) - m_S(f))|S|$$

Since f is bounded, $|f(x)| \leq M$ and hence $\sum_{S\in(\mathbb{S})^1}(M_S(f) - m_S(f))|S| \leq 2M\sum_{S\in(\mathbb{S})^1}|S|$. Since we know that S_1 consists of the subrectangles of the cover elements (and only the cover elements) we note that $\sum_{S\in(\mathbb{S})^1}|S| \leq \varepsilon$. Hence we have the first sum $\leq 2M\varepsilon$

Look at the second sum:

$$\sum_{S\in(\mathbb{S})^2}(M_S(f) - m_S(f))|S|$$

For $S \in (\mathbb{S})^2$, we have $\operatorname{osc}(f,x) < \varepsilon$. By A.2.16 we have that there exists a partition of S, call it P_S such that $\mathcal{U}(f,P_S) - \mathcal{L}(f,P_S) \leq \varepsilon|S|$. Extending the grid for each of these subrectangles in P_S forms a new partition P' which is a refinement of P as a whole, such that $\sum_{S'\subset S;S'\in P'}(M_S(f) - m_S(f))|S'| \leq \varepsilon|S|$, so as a whole,

$$\mathcal{U}(f,P') - \mathcal{L}(f,P') = \sum_{S'\in(\mathbb{S})^1}\sum_{S''\subset S';S''\in P'}(M_{S''}(f) - m_{S''}(f))|S''| +$$

$$\sum_{S'\in(\mathbb{S})^2}\sum_{S''\subset S';S''\in P'}(M_{S''}(f) - m_{S''}(f))|S''|$$

$$\leq 2M\varepsilon + \sum_{S'\in(\mathbb{S})^2}\varepsilon|S'| \leq 2M\varepsilon + \varepsilon|A|$$

Hence f is integrable by criteria A.2.7 we have that f is integrable.

$\Longleftarrow$) Let f be integrable. The set of discontinuities B is $\cup_{n\in\mathbb{N}}B_n$ where $B_n := \{x \in A : \operatorname{osc}(f,x) \geq 1/n\}$ (closed and hence compact). For $a \in B_n$, we have $\operatorname{osc}(f,a) \geq 1/n$ which means that $\inf_{\delta>0}(M_{R_\delta}(f) - m_{R_\delta}(f)) \geq 1/n$. Let P be the partition of A such that

$$\mathcal{U}(f,P) - \mathcal{L}(f,P) \leq \varepsilon/n$$

and let SS be the collection of subrectangles of P which intersect with B_n. These form a cover for B_n. Therefore, for S such that $S \cap B_n \neq \varnothing$, we have $\exists x \in S$ such that $\inf_{\delta>0}(M_{R_\delta}(f) - m_{R_\delta}(f)) \geq 1/n$ which implies $M_S(f) - m_S(f) \geq 1/n$ and hence

$$\varepsilon/n \geq \sum_{S\in SS}(M_S(f) - m_S(f))|S| \geq 1/n\sum_{S\in SS}|S|$$

which gives us that $\sum_{S\in SS}|S| \leq \varepsilon$. Therefore, B_n is content zero and hence B is of measure zero.

∎

Definition A.2.18. ($\int_C f$ for arbitrary set C) Let A be any rectangle containing C (in the case where we can find one). Then we define

$$\int_C f := \int_A f \cdot \chi_C$$

where $\chi_C(x) = \begin{cases} 1 & \text{if x } \in C \\ 0 & \text{if x } \notin C \end{cases}$

Theorem A.2.19. *Function $\chi_C : A \to \mathbb{R}$ is integrable if and only if ∂C is a set of measure zero.*

> *Proof.* Note that, for $x \in C^\circ$, χ_C is continuous. And likewise for $x \in (\mathbb{R}^n - C)^\circ$. The only place where χ_C is discontinuous is precisely on ∂C. The result follows from A.2.17.

■

Definition A.2.20. (Jordan-Measurable) A bounded set C whose boundary has measure zero is called **Jordan Measurable**.

Theorem A.2.21. *If $f : A \to \mathbb{R}$ is a non-negative function and $\int_A f = 0$, then $B = \{x : f(x) > 0\}$ is a set of measure zero.*

> *Proof.* Let $B_n := \{x \in A : f(x) \geq 1/n\}$ and note that $B = \cup_n B_n$. Since f is said to be integrable, we note that $\inf_P \mathcal{U}(f, P)$ must be equal to the integral $\int_A f = 0$. Let $\varepsilon > 0$ be arbitrary and consider that partition P such that $\mathcal{U}(P, f) < \varepsilon/n$. Look at $\mathbb{J} := \{S \in P : S \cap B_n \neq \varnothing\}$. This covers B_n. Moreover, $1/n \sum_{S \in \mathbb{J}} |S| \leq \sum_{S \in \mathbb{J}} M_S(f)|S| \leq \mathcal{U}(f, P) \leq \varepsilon/n$ (Since $f(x) \geq 1/n$ for elements in B_n, and each S has some intersection with B_n). This implies $\sum_{S \in \mathbb{J}} |S| \leq \varepsilon$ and $\cup_{S \in \mathbb{J}} S$ covers B_n. Therefore, B_n is content zero set, and hence B is a measure zero set.

■

A.2.4 Fubini's Theorem

Definition A.2.22. (Lower and Upper Integrals) If $f : A \to \mathbb{R}$ is a bounded function on a closed rectangle $A \subset \mathbb{R}^n$, then regardless of whether f is integrable, the least upper bound of the lower sums and the greatest lower bound of the upper sums both exist. They are called **Lower** and **Upper** *integrals* of f on A, denoted by:

$$\mathbf{L}\int_A f$$

and

$$\mathbf{U}\int_A f$$

Theorem A.2.23. ***(Fubini)*** *Let $A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}^m$ be closed rectangles and let $f : A \times B \to \mathbb{R}$ be integrable. For $x \in A$, let $g_x : B \to R$ be given by $g_x(y) = f(x, y)$ (and likewise define $h_y : A \to \mathbb{R}$ as $h_y(x) = f(x, y)$). Let*

$$\mathscr{L}(x) := \mathbf{L}\int_B g_x = \mathbf{L}\int_B f(x, y)dy$$

and

$$\mathscr{U}(x) := \mathbf{U}\int_B g_x = \mathbf{U}\int_B f(x, y)dy$$

then $\mathscr{L}(x) : A \to \mathbb{R}$ and $\mathscr{U}(x) : A \to \mathbb{R}$ are both integrable *on A and*

$$\int_{A \times B} f = \int_A \mathscr{L}(x)dx = \int_A (\mathbf{L}\int_B f(x, y)dy)dx$$

$$\int_{A \times B} f = \int_A \mathscr{U}(x)dx = \int_A (\mathbf{U}\int_B f(x, y)dy)dx$$

> *Proof.* **(sketch)**

■

A.3 Partitions of Unity

Lemma A.3.1. *There exists a function* $f : \mathbb{R} \to \mathbb{R}$ *which is positive on* $(-1, 1)$*, 0 elsewhere and is* C^∞

Proof. Consider

$$f := \begin{cases} e^{-(x-1)^{-2}} e^{-(x+1)^{-2}} & \text{if } x \in (-1,1) \\ 0 & \text{elsewhere} \end{cases}$$

This function is C^∞ and is positive on $(-1, 1)$ and 0 elsewhere.

■

Lemma A.3.2. *Let* $a \in \mathbb{R}^n$*,* $g : \mathbb{R}^n \to \mathbb{R}$ *be given by*

$$g(x) = f([x^1 - a^1]/\varepsilon) f([x^2 - a^2]/\varepsilon) \cdots f([x^n - a^n]/\varepsilon)$$

g *is then* C^∞*, and has positive values only on* $(a^1 - \varepsilon, a^1 + \varepsilon) \times (a^2 - \varepsilon, a^2 + \varepsilon) \cdots \times (a^n - \varepsilon, a^n + \varepsilon)$.

Appendix B

Differential Forms

B.1 Exterior Algebra: Definitions and Results

Definition B.1.1. (Dual Space V^*) Let V be a vector space over $\mathbb{R}$. The set of all linear maps $f : V \to \mathbb{R}$ is called the *dual space* of V and is denoted by V^* or $\Lambda^1(V)$.

Lemma B.1.2. *Let $e_1, e_2, \cdots, e_n$ be a basis for V. Then the vectors $\xi^1, \xi^2, \cdots, \xi^n \in \Lambda^1(V)$ defined by $\xi^i(e_j) = \delta_{ij}$ forms a basis for $\Lambda^1(V)$.*

Definition B.1.3. (Multilinear Maps (Tensors))

$$f : \underbrace{\mathbb{R}^n \times \mathbb{R}^n \times \cdots \times \mathbb{R}^n}_{k \text{ copies}} \to \mathbb{R}$$

is a *k-tensor* if it is linear in each coordinate.

Note: The space of all tensors forms a vector space.

Definition B.1.4. (σf) Let f be a $k-$ tensor and $\sigma \in Sym(k)$. We define

$$\sigma f(v_1, v_2, \cdots, v_k) := f(v_{\sigma(1)}, v_{\sigma(2)}, \cdots, v_{\sigma(k)})$$

B.1.1 σf as a group action

Let X = set of all $k-$ tensors. Define $\rho : sym(k) \times X \to X$ by $\rho(\sigma, f) := \sigma f$. We see that ρ is a group action since

1. $\rho(1, f) = f$
2. $\rho(\sigma_1, \rho(\sigma_2, f)) = \rho(\sigma_1\sigma_2, f)$

Definition B.1.5. (Alternating Tensors) A $k-$tensor is alternating if for every $\sigma \in sym(k)$,

$$f(v_{\sigma(1)}, v_{\sigma(2)}, \cdots, v_{\sigma(k)}) = \operatorname{sign}(\sigma) f(v_1, v_2, \cdots, v_k)$$

or

$$\sigma f \equiv \operatorname{sign}(\sigma) f$$

Definition B.1.6. (Symmetric Tensors) A $k-$tensor is symmetric if for every $\sigma \in sym(k)$,

$$f(v_{\sigma(1)}, v_{\sigma(2)}, \cdots, v_{\sigma(k)}) = f(v_1, v_2, \cdots, v_k)$$

or

$$\sigma f \equiv f$$

Definition B.1.7. (Symmetrization Operator Sf)

$$S : \{k-\text{tensors}\} \to \{k-\text{tensors}\}$$

given by $Sf(v_1, v_2, \cdots, v_k) = \sum_{\sigma \in sym(k)} f(v_{\sigma(1)}, v_{\sigma(2)}, \cdots, v_{\sigma(k)})$ or

$$Sf \equiv \sum_{\sigma \in sym(k)} \sigma f$$

is called the *symmetrization operator*

Definition B.1.8. (Alternation or Antisymmetrizer Operator Af)

$$A : \{k-\text{tensors}\} \to \{k-\text{tensors}\}$$

given by $Af(v_1, v_2, \cdots, v_k) = \sum_{\sigma \in sym(k)} \operatorname{sign}(\sigma) f(v_{\sigma(1)}, v_{\sigma(2)}, \cdots, v_{\sigma(k)})$ or

$$Af \equiv \sum_{\sigma \in sym(k)} \operatorname{sign}(\sigma) \sigma f$$

is called the *alternation operator*

Theorem B.1.9. *If f is a $k-$tensor, then*

1. *Sf is a symmetric $k-$tensor.*
2. *Af is an alternating $k-$tensor.*

Proof. 1) $Sf := \sum_{\sigma \in sym(k)} \sigma f$. Let $\tau \in sym(k)$.

$$\tau Sf = \sum_{\sigma \in sym(k)} (\tau\sigma) f = Sf$$

(since $\tau\sigma$ runs over all of $sym(k)$ once and exactly once).

2) $Af := \sum_{\sigma \in sym(k)} \operatorname{sign}(\sigma) f$. Let $\tau \in sym(k)$.

$$\tau Af = \sum_{\sigma \in sym(k)} \operatorname{sign}(\sigma\tau) \operatorname{sign}(\tau) (\tau\sigma) f =$$

$$\operatorname{sign}(\tau) Af$$

.

∎

Lemma B.1.10.

1. *If f is an alternating $k-$ tensor ($\sigma f = \operatorname{sign}(\sigma) f$) then $Af = k! f$.*
2. *If f is an alternating $k-$ tensor then $Sf = 0$.*
3. *If f is a symmetric $k-$tensor ($\sigma f = f$) then $Af = 0$.*
4. *If f is a symmetric $k-$tensor then $Sf = k! f$*

Proof. 1)

$$Af := \sum_{\sigma \in sym(k)} \operatorname{sign}(\sigma) \operatorname{sign}(\sigma) f = \sum_{\sigma \in sym(k)} f = k! f$$

2) If f is alternating, then

$$Sf = \sum_{\sigma \in sym(k)} \operatorname{sign}(\sigma) f = 0$$

3) If f is symmetric, then

$$A(f) := \sum_{\sigma \in sym(k)} \operatorname{sign}(\sigma) f = 0$$

4) If f is symmetric, then

$$Sf := \sum_{\sigma \in sym(k)} \sigma f = k! f$$

■

B.1.2 Tensor Product, Alternating Forms and the Wedge Product

Definition B.1.11. (Tensor Product) Let f be a $k-$tensor and g be a $l-$tensor. We define the **tensor product** $f \otimes g$ to be the $(k+l)-$tensor given by

$$f \otimes g(v_1, v_2, \cdots, v_k, v_{k+1}, v_{k+2}, \cdots, v_{k+l}) := f(v_1, v_2, \cdots, v_k) \cdot g(v_{k+1}, v_{k+2}, \cdots, v_{k+l})$$

Note: The tensor product is *bilinear* (linear in each argument) and *associative* ($(f \otimes g) \otimes h = f \otimes (g \otimes h)$) but not *commutative.*

Definition B.1.12. (Wedge Product) Let f be an alternating $k-$tensor ($f \in \Lambda^k(V)$) and g an alternating $l-$tensor ($g \in \Lambda^l(V)$). We define

$$f \wedge g := \frac{1}{k!l!} A(f \otimes g)$$

or written explicitly

$$f \wedge g(v_1, \cdots, v_{k+l}) := \frac{1}{k!l!} \sum_{\sigma \in sym(k+l)} \operatorname{sign}(\sigma) f(v_{\sigma(1)}, \cdots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \cdots, v_{\sigma(k+l)})$$

Definition B.1.13. ($Sym(k,l)$) Given natural numbers k and l, we define $Sym(k,l)$ as follows:

$$Sym(k,l) := \{\sigma \in sym(k+l) : \sigma(1) < \sigma(2) < \cdots < \sigma(k) \text{ and } \sigma(k+1) < \sigma(k+2) < \cdots < \sigma(k+l)\}$$

Theorem B.1.14. ***(Equivalent Expressions for the Wedge Product)*** *The two expressions*

1.

$$f \wedge g(v_1, \cdots, v_{k+l}) := \frac{1}{k!l!} \sum_{\sigma \in sym(k+l)} \operatorname{sign}(\sigma) f(v_{\sigma(1)}, \cdots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \cdots, v_{\sigma(k+l)})$$

2.

$$f \wedge g(v_1, \cdots, v_{l+l}) = \sum_{\sigma \in sym(k,l)} \operatorname{sign}(\sigma) f(v_{\sigma(1)}, \cdots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \cdots, v_{\sigma(k+l)})$$

are equivalent.

Proof. Consider $G = sym(k+l)$ and consider the subgroup

$$H := \{\sigma \in sym(k+l) : \sigma|_{\{1,\cdots,k\}} \equiv \mathbb{1} \text{ and } \sigma|_{\{k+1,\cdots,l+l\}} \equiv \mathbb{1}\}$$

(One can view every element $\sigma \in H$ as $\pi_1 \times \pi_2$ where $\pi_1 \in sym\{1, \cdots, k\}$ and $\pi_2 \in sym\{k+1, \cdots, k_l\}$).

Let $\alpha, \beta \in G$ and define an equivalence relation $\alpha \sim \beta$ if and only if there exists $h \in H$ such that $\alpha h = \beta$

Let S_σ be the expression $\text{sign}(\sigma)\sigma(f \otimes g)$.

Suppose $\alpha \sim \beta$ or $\alpha h = \beta$ where $h = \pi_1 \times \pi_2$.

$$S_\alpha = \text{sign}(\alpha) f(v_{\alpha(1)}, \cdots, v_{\alpha(k)}) g(v_{\alpha(k+1)}, \cdots, v_{\alpha(k+l)})$$

and

$$S_\beta = \text{sign}(\beta) f(v_{\beta(1)}, \cdots, v_{\beta(k)}) g(v_{\beta(k+1)}, \cdots, v_{\beta(k+l)})$$

which is equal to

$$\text{sign}(\alpha)\,\text{sign}(\pi_1)\,\text{sign}(\pi_2) f(v_{\alpha(h(1))}, \cdots, v_{\alpha(h(k))}) g(v_{\alpha(h(k+1))}, \cdots, v_{\alpha(h(k+l))})$$

$$= \text{sign}(\alpha)\,\text{sign}(\pi_1)^2\,\text{sign}(\pi_2)^2 f(v_{\alpha(1)}, \cdots, v_{\alpha(k)}) g(v_{\alpha(k+1)}, \cdots, v_{\alpha(k+l)}) = S_\alpha$$

So whenever $\alpha \sim \beta$, $S_\alpha = S_\beta$.

We can also see that, via rearrangement, for every $\sigma \in sym(k+l)$ there exists an $\alpha \in sym(k,l)$ and a $h \in H$ such that $\sigma h = \alpha$ or $\sigma \sim \alpha$.

The first expression can now be written as

$$\frac{1}{k!l!} \sum_{\sigma \in sym(k,l)} \sum_{\alpha \in [\sigma]} S_\alpha$$

but for elements of the same equivalence class, S_α are all the same and is equal to S_σ, so we have

$$\frac{1}{k!l!} \sum_{\sigma \in sym(k,l)} S_\sigma(k!l!) = \sum_{\sigma \in sym(k,l)} \text{sign}(\sigma)\sigma(f \otimes g)$$

(The $k!l!$ comes from the fact that $|H| = k!l!$ and so the size of each coset is $k!l!$) and the equality is proved.

∎

Theorem B.1.15. *For $f \in \Lambda^k(V)$ and $g \in \Lambda^l(V)$*

$$f \wedge g = (-1)^{kl} g \wedge f$$

Proof.

$$f \wedge g = \sum_{\sigma \in sym(k,l)} \text{sign}(\sigma)\sigma(f \otimes g)$$

Let τ be the element of $sym(k+l)$ given by $\tau(1) = k+1, \tau(2) = k+2, \cdots \tau(l) = k+l$ and $\tau(l+1) = 1, \tau(l+2) = 2, \cdots \tau(l+k) = k$ (We are moving the block $\{k+1, k+2, \cdots, k+l\}$ back, behind the block $\{1, 2 \cdots k\}$, which amounts to a total of kl swaps/transpositions). Note that τ acts as a correspondence between $sym(k,l)$ and $sym(l,k)$ since for every $\sigma \in sym(k,l)$, we have

$\sigma(\tau(l+1)) < \sigma(\tau(l+2)) < \cdots < \sigma(\tau(l+k))$ and $\sigma(\tau(1)) < \sigma(\tau(2)) < \cdots < \sigma(\tau(l))$ so $\sigma(\tau)$ is in $sym(l,k)$.

$$f \wedge g(v_1, \cdots, v_{k+l}) = \sum_{\sigma \in sym(k,l)} \mathrm{sign}(\sigma) f(v_{\sigma(1)}, \cdots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \cdots, v_{\sigma(k+l)})$$

$$= \sum_{\sigma \in sym(k,l)} \mathrm{sign}(\tau\sigma)\,\mathrm{sign}(\tau) g(v_{\sigma(\tau(1))}, \cdots, v_{\sigma(\tau(l))}) f(v_{\sigma(\tau(l+1))}, \cdots, v_{\sigma(\tau(l+k))})$$

$$= \sum_{\gamma \in sym(l,k)} \mathrm{sign}(\gamma)(-1)^{kl} g(v_{\gamma(1)}, \cdots, v_{\gamma(l)}) f(v_{\gamma(l+1)}, \cdots, v_{\gamma(l+k)}) = (-1)^{kl} g \wedge f$$

∎

B.1.3 The Associativity of the Wedge Product

Lemma B.1.16. *Let $f \in \Lambda^k(V)$ and $g \in \Lambda^l(V)$, then:*

1. $A(Af \otimes G) = k!A(f \otimes g)$
2. $A(f \otimes Ag) = l!A(f \otimes g)$

Proof. 1)

$$A(Af \otimes g) = \sum_{\sigma \in sym(k+l)} \mathrm{sign}(\sigma)\sigma(Af \otimes g)$$

$$Af := \sum_{\substack{\tau \in sym(k+l) \\ \tau \text{ preserves } \{k+1, \cdots, k+l\}}} \mathrm{sign}(\tau)\tau(f)$$

and

$$A(Af \otimes g) := \sum_{\sigma \in sym(k+l)} \mathrm{sign}(\sigma)(Af(X_{\sigma(1)}, \cdots, X_{\sigma(k)}) g(X_{\sigma(k+1)}, \cdots, X_{\sigma(k+l)}))$$

$$= \sum_{\sigma \in sym(k+l)} \mathrm{sign}(\sigma) \Big(\sum_{\substack{\tau \in sym(k+l) \\ \tau \text{ preserves} \\ \{k+1, \cdots, k+l\}}} \mathrm{sign}(\tau) f(X_{\sigma\tau(1)}, \cdots, X_{\sigma\tau(k)}) \Big) g(X_{\sigma(k+1)}, \cdots, X_{\sigma(k+l)}))$$

$$= \sum_{\sigma \in sym(k+l)} \sum_{\substack{\tau \in sym(k+l) \\ \tau \text{ fixes} \\ \{k+1, \cdots, k+l\}}} \mathrm{sign}(\sigma\tau) f(X_{\sigma\tau(1)}, \cdots, X_{\sigma\tau(k)}) g(X_{\sigma\tau(k+1)}, \cdots, X_{\sigma\tau(k+l)})$$

The number of such τ that fixes $\{k+1, \cdots, k+l\}$ are precisely $k!$, and for each such τ, $\sigma\tau$ runs over all of $sym(k+l)$ once and exactly once. Hence we have

$$A(Af \otimes g) = k! \sum_{\gamma \in sym(k+l)} \mathrm{sign}(\gamma)\gamma(f \otimes g) = k!A(f \otimes g)$$

2) The 2nd part is proved similarly.

∎

Theorem B.1.17. ***(Associativity of Wedge Product)*** *For any $f \in \Lambda^k(V), g \in \Lambda^l(V), h \in \Lambda^m(V)$*

$$f \wedge (g \wedge h) = (f \wedge g) \wedge h$$

Proof.

$$f \wedge (g \wedge h) := \frac{1}{k!(l+m)!} A(f \otimes (g \wedge h)) = \frac{1}{k!(l+m)!} A(f \otimes (\frac{1}{l!m!} A(g \otimes h)))$$

$$= \frac{1}{k!l!m!(l+m)!} A(f \otimes A(g \otimes h)) = \frac{1}{k!l!m!} A(f \otimes g \otimes h)$$

and likewise

$$(f \wedge g) \wedge h = \frac{1}{(k+l)!m!} A((f \wedge g) \otimes h) = \frac{1}{(k+l)k!l!m!} A(A(f \otimes g) \otimes h)$$

$$= \frac{1}{k!l!m!} (k+l)! A(f \otimes g \otimes h) = \frac{1}{k!l!m!} A(f \otimes g \otimes h)$$

∎

Note: *The proof of B.1.17 shows that if f_{l_1} is an l_1-form, f_{l_2} is an l_2-form and so on, $f_{l_1} \wedge f_{l_2} \wedge \cdots \wedge f_{l_k}$ is given by*

$$f_{l_1} \wedge f_{l_2} \wedge \cdots \wedge f_{l_k} = \frac{1}{l_1!l_2! \cdots l_k!} A(f_{l_1} \otimes f_{l_2} \otimes \cdots \otimes f_{l_k}) \tag{B.1.1}$$

Short Digression on Determinants

Definition B.1.18. (Determinant (*Leibniz Formula*)) Let $\{a_{ij} : 1 \leq i, j \leq n\}$ be a matrix. We define determinant of A as

$$\det(A) := \sum_{\sigma \in sym(n)} \text{sign}(\sigma) a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)}$$

(To be thought of as multiplying one element from each row chosen from distinct columns, and taking into account the sign of the permutation of the columns).

Definition B.1.19. (Minor of $(i,j)-$entry) Let A be a matrix and let a_{ij} be the $(i,j)-$entry of A. The *minor* of a_{ij} is the determinant of the matrix obtained by deleting the i^{th} row and j^{th} column from A. It is denoted m_{ij}. The matrix itself (after deleting i^{th} row and j^{th} column) is denoted M_{ij}.

Definition B.1.20. (Laplace's Expansion) Fix a row i. Then the Laplace's expansion gives

$$\det(B) = \sum_{j=1}^{n} (-1)^{i+j} b_{ij} \det(M_{ij})$$

Note: From

$$\sum_{\sigma \in sym(n)} \text{sign}(\sigma) \prod_{q=1}^{n} a_{q,\sigma(q)} = \sum_{j=1}^{n} \sum_{\substack{\sigma \in sym(n) \\ \sigma : i \mapsto j}} \text{sign}(\sigma) a_{i,j} \prod_{\substack{q=1 \\ q \neq i}}^{n} a_{q,\sigma(q)} \tag{B.1.2}$$

we can see that the Laplace's expansion is a consequence of the Leibniz formula for determinants.

Theorem B.1.21. ***(Wedge Product of** $1-$**forms or Functionals)*** *Let $f_1, f_2, \cdots, f_k$ be in $\Lambda^1(V) \equiv V^*$. Then*

$$f_1 \wedge f_2 \wedge \cdots \wedge f_k(x_1, x_2, \cdots, x_k) = det(f_i(x_j))$$

Proof. From B.1.1, we have

$$f_1 \wedge f_2 \wedge \cdots \wedge f_k(x_1, \cdots, x_k) = A(f_1 \otimes f_2 \otimes \cdots \otimes f_k)$$

or

$$f_1 \wedge f_2 \wedge \cdots \wedge f_k(x_1, x_2, \cdots, x_k) = \sum_{\sigma \in sym(k)} \text{sign}(\sigma) f_1(x_{\sigma(1)}) f_2(x_{\sigma(2)}) \cdots f_k(x_{\sigma(k)})$$

which is precisely the Leibniz formula for determinants. Hence

$$f_1 \wedge f_2 \wedge \cdots \wedge f_k(x_1, x_2, \cdots, x_k) = \det(f_i(x_j))$$

■

Lemma B.1.22. *Any collection $f_1, f_2, \cdots, f_k$ of $1-$forms is linearly independent if and only if $f_1 \wedge f_2 \wedge \cdots \wedge f_k \neq 0$*

Proof. $\implies$) Say $f_1, \cdots, f_k$ is a linearly independent collection of functionals. Then it is part of a larger collection that forms a basis for V^*. Let $x_1, x_2, \cdots, x_k$ be the vectors in V such that $f_i(x_j) = \delta_{ij}$. Then

$$f_1 \wedge f_2 \wedge \cdots \wedge f_k(x_1, x_2, \cdots, x_k) = \det(\delta_{ij}) = 1$$

$\impliedby$) Let $f_1, f_2, \cdots, f_k$ be linearly dependent. Then (without loss of generality) $f_1 = c_2 f_2 + c_3 f_3 + \cdots c_k f_k$ and $f_1 \wedge f_2 \wedge \cdots \wedge f_k = c_2(f_2 \wedge f_2 \wedge f_3 \cdots \wedge f_k) + c_3(f_3 \wedge f_2 \wedge f_3 \wedge \cdots \wedge f_k) + \cdots c_k(f_k \wedge f_2 \wedge f_3 \wedge \cdots \wedge f_k) = 0$

■

B.1.4 Dimension of Alternating k-forms

Let $E_1, E_2, \cdots, E_n$ be a basis for V. Let $\xi^1, \xi^2, \cdots, \xi^n \in V^* = \Lambda^k(V)$ be the dual basis to $E_1, \cdots E_n$.

Theorem B.1.23. ***(Basis for** $\Lambda^k(V)$**)** The set*

$$\{\xi^{i_1} \wedge \xi^{i_2} \wedge \cdots \wedge \xi^{i_k} : 1 \leq i_1 < i_2 < \cdots < i_k \leq n\}$$

forms a basis for $\Lambda^k(V)$.

Proof. **Linear Independence:** Look at

$$\sum_{1 \leq i_1 < \cdots < i_k \leq n} c_{i_1 i_2 \cdots i_k} \xi^{i_1} \wedge xi^{i_2} \wedge \cdots \wedge \xi^{i_k} \equiv 0$$

in $\Lambda^k(V)$. Hence,

$$\sum_{1 \leq i_1 < \cdots < i_k \leq n} c_{i_1 i_2 \cdots i_k} \xi^{i_1} \wedge xi^{i_2} \wedge \cdots \wedge \xi^{i_k}(E_{j_1}, E_{j_2}, \cdots, E_{j_k}) =$$

$$\sum_{1 \leq i_1 < \cdots < i_k \leq n} c_{i_1 i_2 \cdots i_k} det(\xi^{i_r}(E_{j_s})) = 0$$

$$\implies c_{j_1 j_2 \cdots j_k} = 0$$

for every increasing k-tuple $j_1, j_2, \cdots, j_k$ and hence $\{\xi^{i_1} \wedge \xi^{i_2} \wedge \cdots \wedge \xi^{i_k} : 1 \leq i_1 < \cdots < i_k \leq n\}$ is linearly independent.

Span: Let $f \in \Lambda^k(V)$. Let $x_r = \sum_{j_1=1}^{n} \xi^{j_1}(x_r) E_{j_1}$ for $r = 1, 2, \cdots, k$.

$$f(x_1, x_2, \cdots, x_k) = f(\sum_{j_1=1}^{n} \xi^{j_1}(x_1) E_{j_1}, \sum_{j_1=1}^{n} \xi^{j_1}(x_2) E_{j_1}, \cdots, \sum_{j_1=1}^{n} \xi^{j_1}(x_k) E_{j_1})$$

$$= \sum_{j_1=1}^{n} \sum_{j_2=1}^{n} \cdots \sum_{j_k=1}^{n} \xi^{j_1}(x_1)\xi^{j_2}(x_2)\cdots\xi^{j_k}(x_k) f(E_{j_1}, E_{j_2}, \cdots, E_{j_k})$$

If $j_r = j_s$ for some r and s, note that the summand is 0. Hence we rewrite the above sum as:

$$\sum_{\substack{1 \le j_1, j_2, \cdots, j_k \le n \\ j_r \text{ distinct}}} \xi^{j_1}(x_1)\xi^{j_2}(x_2)\cdots\xi^{j_k}(x_k) f(E_{j_1}, E_{j_2}, \cdots, E_{j_k})$$

We now note that each choice of $1 \le j_1, j_2, \cdots, j_k \le n$ tuple is some $\sigma \in sym(k)$ of the ordered $1 \le j_1 < j_2 < \cdots < j_k \le n$ (Simply reorder the elements $j_1, j_2, \cdots, j_k$ such that we achieve $\sigma(j_1) < \sigma(j_2) < \cdots < \sigma(j_k)$), and hence the above sum can be rewritten as:

$$\sum_{1 \le j_1 < \cdots < j_k \le n} \sum_{\sigma \in sym(k)} \xi^{j_{\sigma(1)}}(x_1)\xi^{j_{\sigma(2)}}(x_2)\cdots\xi^{j_{\sigma(k)}}(x_k) f(E_{j_{\sigma(1)}}, \cdots, E_{j_{\sigma(k)}})$$

which is precisely

$$\sum_{1 \le j_1 < \cdots < j_k \le n} f(E_{j_1}, \cdots, E_{j_k}) \sum_{\sigma \in sym(k)} \text{sign}(\sigma)\xi^{j_{\sigma(1)}}(x_1)\xi^{j_{\sigma(2)}}(x_2)\cdots\xi^{j_{\sigma(k)}}(x_k)$$

$$= \sum_{1 \le j_1 < \cdots < j_k \le n} f(E_{j_1}, \cdots, E_{j_k})\xi^{j_1} \wedge \xi^{j_2} \cdots \wedge \xi^{j_k}(x_1, x_2, \cdots, x_k)$$

which means f can be written as a linear combination of $\xi^{j_1} \wedge \cdots \wedge \xi^{j_k}$.

∎

Lemma B.1.24. *Let $\beta^i \in \Lambda^1(V)$ for $1 \le i \le k$ and let $\gamma^i \in \Lambda^1(V)$ and if*

$$\beta^i = \sum_{j=1}^{k} \alpha^i_j \gamma^j$$

then

$$\beta^1 \wedge \beta^2 \wedge \cdots \wedge \beta^k = \det(\alpha^i_j)\gamma^1 \wedge \gamma^2 \wedge \cdots \gamma^k$$

Proof.

$$\beta^1 \wedge \cdots \wedge \beta^k = \sum_{j_1=1}^{k} \sum_{j_2=1}^{k} \cdots \sum_{j_k=1}^{k} (\alpha^1_{j_1}\alpha^2_{j_2}\cdots\alpha^k_{j_k})(\gamma^{j_1} \wedge \gamma^{j_2} \wedge \cdots \wedge \gamma^{j_k})$$

$$= \sum_{\substack{1 \le j_1 < \cdots < j_k \le k \\ j_r \text{ distinct}}} (\alpha^1_{j_1}\alpha^2_{j_2}\cdots\alpha^k_{j_k})\gamma^{j_1} \wedge \gamma^{j_2} \cdots \wedge \gamma^{j_k}$$

$$= \sum_{1 \le j_1 < \cdots < j_k \le k} \sum_{\sigma \in sym(k)} \text{sign}(\sigma)\alpha^1_{j_1} \cdots \alpha^k_{j_k}\gamma^1 \wedge \gamma^2 \cdots \gamma^k$$

$$= \det(\alpha^i_j)\gamma^1 \wedge \gamma^2 \wedge \cdots \wedge \gamma^k$$

∎

Lemma B.1.25. *Let $f \in \Lambda^k(V)$ and let $u_1, u_2, \cdots, u_k$ and $v_1, v_2, \cdots, v_k \in V$ be related by*

$$u_i = \sum_{j=1}^{n} a^i_j v_j$$

Then

$$f(u_1, u_2, \cdots, u_k) = \det(a^i_j) f(v_1, v_2, \cdots, v_k)$$

Proof. $f(\sum_{j_1=1}^{n} a^1_{j_1} v_{j_1}, \sum_{j_2=1}^{k} a^2_{j_2} v_{j_k}, \cdots, \sum_{j_k=1}^{n} a^k_{j_k} v_j)$

$$= \sum_{\substack{j_1, j_2, \cdots, j_k \\ eachdistinct}} a^1_{j_1} a^2_{j_2} \cdots a^k_{j_k} f(v_{j_1}, v_{j_2}, \cdots, v_{j_k})$$

$$= \sum_{\substack{1 \le j_1 < j_2 < \cdots < j_k \le k \\ eachdistinct}} \sum_{\sigma \in sym(k)} \text{sign}(\sigma) a^1_{\sigma(1)} a^2(\sigma(2)) \cdots a^k_{\sigma(k)} f(v_1, v_2, \cdots, v_k) =$$

$$\det(a^i_j) f(v_1, v_2, \cdots, v_k)$$

■

Notation: Let $\mathscr{T}_k$ denote the set

$$\mathscr{T}_k := \{I \equiv (i_1, i_2, \cdots, i_k) : 1 \le i_1 < i_2 < \cdots < i_k \le n\} \tag{B.1.3}$$

Suppose that $E_1, E_2, \cdots, E_n$ are the basis for V, and $\xi^1, \xi^2, \cdots, \xi^n$ be the dual basis for $\Lambda^1(V)$. Then, for $I = (i_1, i_2, \cdots, i_k)$ we have the notation

$$\xi^I := \xi^{i_1} \wedge \xi^{i_2} \wedge \cdots \wedge \xi^{i_k} \tag{B.1.4}$$

and hence for any $k-$form $f : V \times V \times \cdots \times V \to \mathbb{R}$ we have the notation for the expansion

$$f := \sum_{I \in \mathscr{T}_k} f^I \xi^I \tag{B.1.5}$$

B.2 Tangent vectors in $\mathbb{R}^n$ and Vector Fields

We denote points in $\mathbb{R}^n$ by

$$p = (p^1, p^2, \cdots, p^n)$$

and points in $T_p(\mathbb{R}^n)$, the tangent space of $\mathbb{R}^n$ at p (the space of all vectors v eminating from p), by

$$v = [v^1, v^2, \cdots, v^n]^T$$

B.2.1 The Directional Derivative and Derivations

Definition B.2.1. (Directional Derivative) Let $p \in \mathbb{R}^n$ be given and let $f \in C^\infty_p$, the space of all functions $f : U \to \mathbb{R}$ such that $p \in U$ and f is C^∞ in U. Let $v \in T_p(\mathbb{R}^n)$. Let $\gamma(t) := p + tv$ and look at

$$\frac{d}{dt}|_{t=0}(f \circ \gamma(t)) = Df(\gamma(0)) \cdot \gamma'(0) = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}|_p f$$

If we fix a point p (arbitrary) and a direction $v \in T_p(\mathbb{R}^n)$, we have an operator $D_{p,v} : C^\infty_p \to \mathbb{R}$ given by

$$D_{p,v}(f) := \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\bigg|_p (f)$$

This operator is *Linear* and follows the property

$$D_{p,v}(f \cdot g) = (D_{p,v}(f))g(p) + (D_{p,v}(g))f(p)$$

*The operator $D_{p,v}$ is called the **directional derivative** at p in the direction v.*

Definition B.2.2. (Germs of Functions) Fix a point p and look at all pairs (f, U) where $f : U \to \mathbb{R}$, $p \in U \subseteq \mathbb{R}^n$ open and $f \in C^\infty(U)$. We say $(f, U) \sim (g, V)$ if f and g agree on an open subset of $U \cap V$. This is an equivalence relation and we (by abuse of notation) denote the equivalence class of (f, U) as simply f. The set of all such equivalence classes is denoted by C_p^∞ known as the ***Germs of functions at*** p.

Defining addition, scalar multiplication and function multiplication on C_p^∞ by the usual operations on representatives, we see that C_p^∞ is a commutative ring with identity. Moreover, it is also a Vector space over $\mathbb{R}$.

Definition B.2.3. (Point Derivation) Fix a point $p \in \mathbb{R}^n$. We say $D : C_p^\infty \to \mathbb{R}$ is a *point derivation* (at p) if:

1. D is linear.
2. D obeys *Leibniz's Rule*:
$$D(f \cdot g) = D(f)g(p) + D(g)f(p)$$

and the space of all point derivations at p is denoted by $\mathscr{D}_p(\mathbb{R}^n)$

Lemma B.2.4. *If $D \in \mathscr{D}_p(\mathbb{R}^n)$, then $D(c) = 0$ for every constant function $c \in C_p^\infty$.*

Proof. $D(1) = D(1 \cdot 1) = 1D(1) + 1D(1) = 2D(1)$ which implies $D(1) = 0$. From linearity, we have the result.

■

Lemma B.2.5. *The space of all point derivations $\mathscr{D}_p(\mathbb{R}^n)$ is a vector space over $\mathbb{R}$ and $T_p(\mathbb{R}^n) \cong \mathscr{D}_p(\mathbb{R}^n)$. Moreover, $D_{p,()} : T_p(\mathbb{R}^n) \to \mathscr{D}_p(\mathbb{R}^n)$ given by $v \mapsto D_{p,v}$ is the isomorphism.*

Proof. That $\mathscr{D}_p(\mathbb{R}^n)$ is a vector space is clear. Look at the map $v \mapsto D_{p,v}$. $v + cw \mapsto D_{p,v+cw}$ Where
$$D_{p,v+cw}(f) = \sum_{i=1}^{n} (v^i + cw^i) \frac{\partial}{\partial x^i}\Big|_p (f) = D_{p,v}(f) + cD_{p,w}(f)$$
Which makes $v \mapsto D_{p,v}$ linear.

Injectivity: Let $D_{p,v}(f) = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\big|_p (f) \equiv 0$ for all $f \in C_p^\infty$ Let $f = x^j : \mathbb{R}^n \to \mathbb{R}$ that picks out the $j-$th coordinate. Then $D_{p,v}(x^j) = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\big|_p (x^j) = v^j = 0$. Therefore, $v = 0$ and hence, the nullspace is trivial.

Surjectivity: Let $f \in C_p^\infty$. We have $f(x) = f(p) + \sum_{i=1}^{n} g_i(x)(x^i - p^i)$ where $g^i(p) = \frac{\partial f}{\partial x^i}(p)$ (From A.1.1)

Let $D \in \mathscr{D}_p(\mathbb{R}^n)$.
$$D(f(x)) = D(f(p)) + \sum_{i=1}^{n} D(g_i(x) \cdot (x^i - p^i))$$
$$= 0 + \sum_{i=1}^{n} D(g_i(x))(0) + D(x^i - p^i) \frac{\partial f}{\partial x^i}(p) = \sum_{i=1}^{n} D(x^i) \frac{\partial}{\partial x^i}\Big|_p f$$
Which means
$$D \equiv \sum_{i=1}^{n} D(x^i) \frac{\partial}{\partial x^i}\Big|_p \equiv D_{p,<D(x^1),D(x^2)\cdots,D(x^n)>}$$

And hence, the map $v \mapsto D_{p,v}$ is surjective.

■

Definition B.2.6. (Vector Field) A *vector field* on $\mathbb{R}^n$ is a map $X : U \mapsto \amalg_{p\in U} T_p(U)$. For every $p \in U$, $X(p)$ is a vector in the tangent space at p.

$$X(p) = \sum_{i=1}^{n} a^i(p) \frac{\partial}{\partial x^i}\Big|_p$$

We call X a smooth vector field if $a^i : U \to \mathbb{R}$ are C^∞.

Let $f \in C_p^\infty$, that is $f : U \to \mathbb{R}$, a vector field X to f is

$$Xf(p) := \sum a_i(p) \frac{\partial}{\partial x^i}|_p f$$

Then

$$Xf : U \to \mathbb{R}$$

is again a C^∞ function.

B.3 Differential Forms

We start with defining what are 1−forms.

Definition B.3.1. (1−forms) A function

$$\omega : U \to \amalg_{p\in U} \Lambda^1(T_p(\mathbb{R}^n))$$

such that $\omega(p) \in \Lambda^1(T_p(\mathbb{R}^n)) \equiv T_p^*(\mathbb{R}^n)$ is called a 1−form.

Note: Let $f : U \to \mathbb{R}$ be any C^∞ function. We can construct a 1−form from f as follows:

$$df : U \to \amalg_{p\in U} \Lambda^1(T_p(U))$$

such that

$$df_p \in \Lambda^1(T_p(U))$$
$$df_p : V \to \mathbb{R}$$

given by the action (for $v = \sum_{j=1}^n v^j \frac{\partial}{\partial x^j}|_p \in T_p(U)$):

$$df_p(v) = df_p(\sum_j v^j \frac{\partial}{\partial x^j}|_p) = \sum_j v^j \frac{\partial}{\partial x^j}|_p(f) \in \mathbb{R}$$

Suppose that $v \in T_p(\mathbb{R}^n)$ and $f \in C_p^\infty(\mathbb{R}^n)$.

$$v = \sum_{i=1}^{n} a^i(p) \frac{\partial}{\partial x^i}\Big|_p \equiv X(p)$$

where $X : U \to \amalg_{p\in U} T_p(\mathbb{R}^n)$ is formally

$$X(p) := \sum_{i=1}^{n} a^i(p) \frac{\partial}{\partial x^i}\Big|_p \in T_p(\mathbb{R}^n)$$

We then have a bilinear function

$$\langle , \rangle : T_p(\mathbb{R}^n) \times C_p^\infty(\mathbb{R}^n) \to \mathbb{R}$$

given by

$$\langle v, f\rangle = \langle \sum_i a^i(p) \frac{\partial}{\partial x^i}\Big|p, f\rangle := \sum_i a^i(p) \frac{\partial}{\partial x^i}\Big|_p f \in \mathbb{R}$$

One can then think of *tangent vectors* as $\langle v, ()\rangle$ functions of the 2nd argument. Likewise, we can think of df as maps on the 1st argument: $df_p() = \langle (), f\rangle$ that takes in tangent vectors and gives a number in $\mathbb{R}$.

Lemma B.3.2. *Let $x^1, x^2, \cdots, x^n : \mathbb{R}^n \to \mathbb{R}$ be the standard coordinate functions on $\mathbb{R}^n$. Then at each point p, $\{dx^1_p, dx^2_p \cdots dx^n_p\}$ forms a basis for $\Lambda^1(T_p(\mathbb{R}^n))$ dual to $\{\frac{\partial}{\partial x^1}\big|_p, \frac{\partial}{\partial x^2}\big|_p, \cdots, \frac{\partial}{\partial x^n}\big|_p\} \in T_p(\mathbb{R}^n)$.*

Proof. Note that

$$dx^i : \mathbb{R}^n \to \cup_{p\in\mathbb{R}^n}\Lambda^1(T_p(\mathbb{R}^n))$$

where $dx^i_p \in \Lambda^1(T_p(\mathbb{R}^n))$ is such that

$$dx^i_p(\frac{\partial}{\partial x^j}\big|_p) := \frac{\partial}{\partial x^j}\big|_p(x^i) = \delta_{ij}$$

and $dx^j_p \in \Lambda_p(T_p(\mathbb{R}^n))$ is precisely the dual basis to $\frac{\partial}{\partial x^i}\big|_p \in T_P(\mathbb{R}^n)$.

■

Note: If

$$\omega : U \to \amalg_{p\in U}\Lambda^1(T_p(\mathbb{R}^n))$$

is a 1−form, $\omega(p) \in \Lambda^1(T_p(\mathbb{R}^n))$ and

$$\omega(p) = \sum_i a^i(p)dx^i_p$$

and formally

$$\omega(\cdot) \equiv \sum_i a_i(\cdot)dx^i_{(\cdot)}$$

Where $a^j(p) = w_p(\frac{\partial}{\partial x^j}\big|_p)$.

B.3.1 Differential Forms in terms of Coordinates

Theorem B.3.3. *Let $f : U \to \mathbb{R}$ be C^∞. Then df_p is defined as that linear function on $T_p(\mathbb{R}^n)$ such that*

$$df_p(\frac{\partial}{\partial x^i}\big|_p) = \frac{\partial}{\partial x^i}\big|_p f$$

Then (as a covector)

$$df_p \equiv \sum_{i=1}^n \frac{\partial f}{\partial x^i}\big|_p (dx^i)_p$$

and formally (as a map on U)

$$df_{()} \equiv \sum_{j=1}^n \frac{\partial f}{\partial x^i}\big|_{()} dx^i_{()}$$

Proof. Let $f : U \to \mathbb{R}$. $df_p \in \Lambda^1(T_p(U))$ and hence we have an expansion of the form

$$df_p = \sum_{j=1}^n a^j_p dx^j_p \tag{B.3.1}$$

where $x^j : \mathbb{R}^n \to \mathbb{R}$ is the j−th standard coordinate function and dx^j is the dual covector to $\frac{\partial}{\partial x^i}\big|_p \in T_p(U)$. Applying df_p on $\frac{\partial}{\partial x^i}\big|_p$ (in B.3.1) we get

$$df_p(\frac{\partial}{\partial x^i}\big|_p) := \frac{\partial f}{\partial x^i}\big|_p = \sum_{j=1}^n a^j_p dx^j_p(\frac{\partial}{\partial x^i}\big|_p) = a^j_p$$

and hence

$$df_p \equiv \sum_{j=1}^n \frac{\partial f}{\partial x^j}\big|_p dx^j_p$$

■

Note: Let $X_p \in T_p(\mathbb{R}^n)$ be a tangent vector. we have $X_p \equiv \sum_{i=1}^n b^i(X_p)\frac{\partial}{\partial x^i}\big|_p$. Apply dx_p^j on both sides:

$$dx_p^j(X_p) = b^j(X_p)$$

and hence

$$X_p \equiv \sum_{i=1}^n dx_p^i(X_p)\frac{\partial}{\partial x^i}\big|_p$$

B.3.2 Differential $k-$forms

Definition B.3.4. (Differential $k-$form) A differential $k-$form is a function of the kind

$$\omega : U \to \amalg_{p\in U}\Lambda^k(T_p(\mathbb{R}^n))$$

$$\omega_p \in \Lambda^k(T_p(\mathbb{R}^n))$$

Since $\Lambda^k(T_p(\mathbb{R}^n))$ is spanned by vectors of the kind (for $1 \le i_1 < i_2 < \cdots < i_k \le n$)

$$dx_p^{i_1} \wedge dx_p^{i_2} \wedge \cdots \wedge dx_p^{i_k}$$

and hence

$$\omega_p \equiv \sum_{1\le i_1<\cdots<i_k\le n} w_{(p)}^{i_1 i_2\cdots i_k} dx_p^{i_1} \wedge dx_p^{i_2} \wedge \cdots \wedge dx_p^{i_k}$$

where $w^{i_1 i_2\cdots i_k} : U \to \mathbb{R}$ is a C^∞ function. Then we say ω is a C^∞ differential $k-$form.

Example B.3.5. (In $\mathbb{R}^3$) Let $x, y, z : \mathbb{R}^3 \to \mathbb{R}$ be the coordinate maps.

1. C^∞ $1-$forms are maps

$$\omega : U \to \amalg_{p\in U}\Lambda^1(T_p(U))$$

and

$$\omega_p \equiv \sum_{i=1}^3 f_p^i dx_p^i = f_p^x dx_p + f_p^y dy_p + f_p^z dz_p \in \Lambda^1(T_p(U))$$

2. C^∞ $2-$forms are maps

$$\omega : U \to \amalg_{p\in U}\Lambda^2(T_p(U))$$

and

$$\omega_p \equiv \sum_{1\le i_1<i_2\le 3} f_p^{i_1 i_2} dx_p^{i_1} \wedge dx_p^{i_2} = f_p^{1,2} dx_p \wedge dy_p - f_p^{1,3} dz_p \wedge dx_p + f_p^{2,3} dy_p \wedge dz_p \in \Lambda^2(T_p(U))$$

3. C^∞ $3-$forms are maps

$$\omega : U \mapsto \amalg_p\Lambda^3(T_p(U))$$

and

$$\omega_p \equiv \sum_{1\le i_1<i_2<i_3} f_p^{i_1 i_2 i_3} dx_p^{i_1} \wedge dx_p^{i_2} \wedge dx_p^{i_3} = f_p dx_p \wedge dy_p \wedge dz_p \in \Lambda^3(T_p(U))$$

B.3.3 Exterior Derivative

Consider a C^∞ function $f : U \to \mathbb{R}$. We have $df : U \to \amalg_{p\in U}\Lambda^1(T_p(U))$ such that $df_p \in \Lambda^1(T_p(U))$.

$$df_p : T_p(U) \to \mathbb{R}$$

via

$$df_p(v) = df_p(\sum_{i=1}^n dx_p^i(v)\frac{\partial}{\partial x^i}\big|_p) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}\big|_p dx_p^i(v)$$

and symbolically we have

$$df \equiv \sum_{i=1}^n \frac{\partial f}{\partial x^i}\big| dx^i$$

So d in this case can be viewed as a map that takes $0-$forms (functions) and gives us a $1-$form df. Before we define the general exterior derivative for $k-$ forms, we will set up some notation.

Definition B.3.6. (The space of $k-$ forms) Let Ω be an open set in $\mathbb{R}^n$. The space of $k-$forms is denoted by

$$\{\Omega; \Lambda^k(T\Omega)\}$$

We also write the space of $k-$ forms as

$$\Omega(\Lambda^k)$$

Definition B.3.7. (Exterior derivative d) Let $\omega \in \{\Omega; \Lambda^k(T\Omega)\}$ be a $k-$form given by (see B.1.3)

$$\omega_p \equiv \sum_{I \in \mathscr{T}_k} \omega_p^I dx_p^I$$

$d : \{\Omega; \Lambda^k(T\Omega)\} \to \{\Omega; \Lambda^{k+1}\Omega\}$ is the function given by (formally)

$$d\omega_p := \big(\sum_{i=1}^{n} \frac{\partial}{\partial x^i}\big|_p dx_p^i\big) \wedge \omega_p \tag{B.3.2}$$

which is

$$\sum_{i=1}^{n} \sum_{I \in \mathscr{T}_k} \frac{\partial \omega^I}{\partial x^i}\big|_p dx_p^i \wedge dx_p^I \tag{B.3.3}$$

Definition B.3.8. (Antiderivation) Let $A = \oplus_{k=0}^{\infty} A^k$ be a graded algebra over a field K. An antiderivation is a map $D : A \to A$ such that

1. D is $K-$linear
2. For all $a \in A^k$, $b \in A^l$,
$$D(ab) = (D(a))b + (-1)^k a(D(b))$$

D is said to be an antiderivation of degree m if $D|_{A^k} : A^k \to A^{k+m}$

Theorem B.3.9. *$d : \oplus_{k=0}^{\infty}\Omega(\Lambda^k) \to \oplus_{k=0}^{\infty}\Omega(\Lambda^k)$ is an antiderivation of degree 1 and*

$$d(\omega \wedge \tau) = d\omega \wedge \tau + (-1)^k \omega \wedge d\tau$$

for all $\omega \in \Omega(\Lambda^k)$ and $\tau \in \Omega(\Lambda^l)$

Proof. That d is linear is clear from formula B.3.3. Let $\omega \equiv \sum_{I \in \mathscr{T}_k} w^I dx^I$ and $\tau \equiv \sum_{J \in \mathscr{T}_l} \tau^J dx^J$. $\omega \wedge \tau \equiv \sum_{I \in \mathscr{T}_k} \sum_{J \in \mathscr{T}_l} w^I \tau^J dx^I \wedge dx^J$ Applying d gives us

$$d(\omega \wedge \tau) = \big(\frac{\partial}{\partial x^i}\big|_p dx_p^i\big) \wedge \big(\sum_{I \in \mathscr{T}_k} \sum_{J \in \mathscr{T}_l} w_p^I \tau_p^J dx_p^I \wedge dx_p^J\big) =$$

$$\sum_{i=1}^{n} \sum_{I \in \mathscr{T}_k} \sum_{J \in \mathscr{T}_l} \left(\frac{\partial \omega^I}{\partial x^i}\big|_p \tau_p^J + \omega_p^I \frac{\partial \tau^J}{\partial x^i}\big|_p\right) dx_p^i \wedge dx_p^I \wedge dx_p^J$$

$$= d\omega \wedge \tau + \sum_{i=1}^{n} \sum_{I \in \mathscr{T}_k} \sum_{J \in \mathscr{T}_l} \omega_p^I \frac{\partial \tau^J}{\partial x^i}\big|_p dx_p^i \wedge dx_p^I \wedge dx_p^J$$

$$= d\omega \wedge \tau + \sum_{i=1}^{n} \sum_{I \in \mathscr{T}_k} \sum_{J \in \mathscr{T}_l} (-1)^k \omega_p^I \frac{\partial \tau^J}{\partial x^i}\big|_p dx_p^I \wedge dx_p^i \wedge dx_p^J$$

$$= d\omega \wedge \tau + (-1)^k \omega \wedge d\tau$$

∎

Lemma B.3.10. $d^2 = 0$

Proof.

$$d\omega = \sum_{i=1}^{n} \sum_{I \in \mathscr{I}_k} \frac{\partial \omega^I}{\partial x^i} dx^i \wedge dx^I$$

and

$$d(d\omega) = \big(\sum_{i=1}^{n} \frac{\partial}{\partial x^i} dx^i \big) \wedge \big(\sum_{j=1}^{n} \sum_{I \in \mathscr{I}_k} \frac{\partial \omega^I}{\partial x^j} dx^j \wedge dx^I \big)$$
$$= \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{I \in \mathscr{I}_k} \frac{\partial^2 \omega^I}{\partial x^i x^j} dx^i \wedge dx^j \wedge dx^I$$

Since $dx^i \wedge dx^j = -dx^j \wedge dx^i$ and $\frac{\partial^2 \omega^I}{\partial x^i x^j} = \frac{\partial^2 \omega^I}{\partial x^j x^i}$ we have that $d(d\omega) = 0$.

■

B.3.4 Closed and Exact Forms

Definition B.3.11. (Closed Form) Let $\omega \in \Omega(\Lambda^k)$ be a $k-$form. We say ω is *closed* if $d\omega = 0$.

Definition B.3.12. (Exact Form) We say $k-$form $\omega \in \Omega(\Lambda^k)$ is *exact* if there exists $\lambda \in \Omega(\Lambda^{k-1})$ such that $d\lambda = \omega$.

Lemma B.3.13. Exact *forms are* closed*.*

B.4 Vector Calculus

We set $n = 3$, $U \subset \mathbb{R}^3$ and $k \in \{1, 2, 3\}$.

1. **One forms:** $\omega : U \to \amalg_{p \in U} T_p^*(\mathbb{R}^3)$ with $\omega_{(p)} \in \Lambda^1(T_p(\mathbb{R}^3)) \equiv T_p^*(\mathbb{R}^3)$ given by

$$\omega_{(p)} \equiv \omega^x_{(p)} dx_{(p)} + \omega^y_{(p)} dy_{(p)} + \omega^z_{(p)} dz_{(p)}$$

2. **Two forms:** $\omega : U \to \amalg_{p \in U} \Lambda^2(T_p(\mathbb{R}^3))$ with $\omega_{(p)} \in \Lambda^2(T_p(\mathbb{R}^3))$ given by

$$\omega_{(p)} \equiv \sum_{1 \le i_1 < i_2 \le 3} \omega^{i_1 i_2}_{(p)} dx^{i_1}_{(p)} \wedge dx^{i_2}_{(p)}$$

$$= \omega^z_{(p)} dx_{(p)} \wedge dy_{(p)} + \omega^x_{(p)} dy_{(p)} \wedge dz_{(p)} + \omega^y_{(p)} dz_{(p)} \wedge dx_{(p)}$$

3. **Three forms:** $\omega : U \to \amalg_{p \in U} \Lambda^3(T_p(\mathbb{R}^3))$ with $\omega_{(p)} \in \Lambda^3(T_p(\mathbb{R}^3))$ given by

$$\omega_{(p)} = w_{(p)} dx_{(p)} \wedge dy_{(p)} \wedge dz_{(p)}$$

We immediately see a correspondence in table B.1

Table B.1: Correspondence between vector fields and forms.

Field	Form
Vector field: $[\omega^x, \omega^y, \omega^z]$	$1-$form: $\omega^x_{(p)} dx_{(p)} + \omega^y_{(p)} dy_{(p)} + \omega^z_{(p)} dz_{(p)}$
Vector field: $[\omega^x, \omega^y, \omega^z]$	$2-$form: $\omega^x_{(p)} dy_{(p)} \wedge dz_{(p)} + \omega^y_{(p)} dz_{(p)} \wedge dx_{(p)} + \omega^z_{(p)} dx_{(p)} \wedge dy_{(p)}$
Scalar field: ω	$3-$form: $\omega_{(p)} dx_{(p)} \wedge dy_{(p)} \wedge dz_{(p)}$

B.4.1 Exterior Derivative of forms in $\mathbb{R}^3$

$0-$forms in $\mathbb{R}^3$

Let $f : U \to \mathbb{R}$. We know that

$$df : U \to \amalg_{p\in U}\Lambda^1(T_p(U))$$

given by

$$df_{(p)} = \frac{\partial f}{\partial x}\Big|_{(p)} dx_{(p)} + \frac{\partial}{\partial y}\Big|_{(p)} dy_{(p)} + \frac{\partial}{\partial z}\Big|_{(p)} dz_{(p)}$$

Which, from the correspondence B.1, can be identified with the vector field

$$\left[\frac{\partial f}{\partial x}\Big|_{(p)}, \frac{\partial f}{\partial y}\Big|_{(p)}, \frac{\partial f}{\partial z}\Big|_{(p)}\right] \equiv \nabla f(p)$$

$1-$forms in $\mathbb{R}^3$

Let $\omega \in \Omega(\Lambda^1(T\mathbb{R}^3))$ be a $1-$form, i.e.

$$\omega_{(p)} \equiv \omega^x_{(p)} dx_{(p)} + \omega^y_{(p)} dy_{(p)} + \omega^z_{(p)} dz_{(p)}$$

$$\text{(Symbolically) } d\omega_p = \left(\sum_{i=1}^{3} \frac{\partial}{\partial x^i}\Big|_p dx^i_p\right) \wedge \omega^x_{(p)} dx_{(p)} + \omega^y_{(p)} dy_{(p)} + \omega^z_{(p)} dz_{(p)}$$

$$= \left(\frac{\partial}{\partial x}\Big|_p dx_p + \frac{\partial}{\partial y}\Big|_p dy_p + \frac{\partial}{\partial z}\Big|_p dz_p\right) \wedge \omega^x_{(p)} dx_{(p)} + \omega^y_{(p)} dy_{(p)} + \omega^z_{(p)} dz_{(p)}$$

$$= \left[\frac{\partial \omega^y}{\partial x}\Big|_p dx_p \wedge dy_p + \frac{\partial \omega^z}{\partial x}\Big|_p dx_p \wedge dz_p\right] + \left[\frac{\partial \omega^x}{\partial y}\Big|_p dy_p \wedge dx_p + \frac{\partial \omega^z}{\partial y}\Big|_p dy_p \wedge dz_p\right] + \left[\frac{\partial \omega^x}{\partial z}\Big|_p dz_p \wedge dx_p + \frac{\partial \omega^y}{\partial z}\Big|_p dz_p \wedge dy_p\right] =$$

$$\left(\frac{\partial \omega^z}{\partial y}\Big|_p - \frac{\partial \omega^y}{\partial z}\Big|_p\right) dy_p \wedge dz_p + \left(\frac{\partial \omega^y}{\partial x}\Big|_p - \frac{\partial \omega^x}{\partial y}\Big|_p\right) dx_p \wedge dy_p + \left(\frac{\partial \omega^x}{\partial z}\Big|_p - \frac{\partial \omega^z}{\partial x}\Big|_p\right) dz_p \wedge dx_p$$

which from the correspondence B.1 can be identified with the vector field

$$\left[\left(\frac{\partial \omega^z}{\partial y}\Big|_p - \frac{\partial \omega^y}{\partial z}\Big|_p\right), \left(\frac{\partial \omega^x}{\partial z}\Big|_p - \frac{\partial \omega^z}{\partial x}\Big|_p\right), \left(\frac{\partial \omega^y}{\partial x}\Big|_p - \frac{\partial \omega^x}{\partial y}\Big|_p\right)\right] \equiv \nabla \times \omega(p)$$

$2-$forms in $\mathbb{R}^3$

Let $\omega \in \Omega(\Lambda^2(T\mathbb{R}^3))$ be a $2-$form i.e.

$$\omega_p \equiv \omega^x_p dy_p \wedge dz_p + \omega^y_p dz_p \wedge dx_p + \omega^z_p dx_p \wedge dy_p \in \Lambda^2(T_p(\mathbb{R}^3))$$

$$d(\omega^x dy \wedge dz)_p = \frac{\partial \omega^x}{\partial x}\Big|_p dx_p \wedge dy_p \wedge dz_p$$

$$d(\omega^y dz \wedge dx)_p = \frac{\partial \omega^y}{\partial y}\Big|_p dy_p \wedge dz_p \wedge dx_p$$

$$d(\omega^z dx \wedge dy)_p = \frac{\partial \omega^z}{\partial z}\Big|_p dz_p \wedge dx_p \wedge dy_p$$

and hence

$$d\omega_p \equiv \left(\frac{\partial \omega^x}{\partial x}\Big|_p + \frac{\partial \omega^y}{\partial y}\Big|_p + \frac{\partial \omega^z}{\partial z}\Big|_p\right) dx_p \wedge dy_p \wedge dz_p$$

Which from the correspondence B.1 can be identified with the scalar field

$$\left.\frac{\partial \omega^x}{\partial x}\right|_p + \left.\frac{\partial \omega^y}{\partial y}\right|_p + \left.\frac{\partial \omega^z}{\partial z}\right|_p \equiv \nabla \cdot \omega(p)$$

From these correspondences, we see that $d : \Omega(\Lambda^0) \to \Omega(\Lambda^1)$ is the gradient, $d : \Omega(\Lambda^1) \to \Omega(\Lambda^2)$ the curl and $d : \Omega(\Lambda^2) \to \Omega(\Lambda^3)$ is the divergence. From this we can see immediately that the well known identities

$$\nabla \times (\nabla f) = 0$$

and

$$\nabla \cdot (\nabla \times \omega) = 0$$

are just special cases of $d^2 = 0$.

B.5 Hodge Star

Let f be a $k-$form given by

$$f \equiv \sum_{I \in \mathscr{T}_k} f^I dx^I$$

We define the operation *Hodge Star* as follows (defined first on basic $k-$forms, and then extended linearly for any $k-$form):

$$\star\, dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k} = \mathrm{sign}(R(I,J)) dx^{j_1} \wedge dx^{j_2} \wedge \cdots \wedge dx^{j_{n-k}}$$

where

1. $I \equiv \{i_1, i_2, \cdots, i_k\} \in \mathscr{T}_k$ and $J \equiv \{j_1, j_2, \cdots, j_{n-k}\}$ being the complementary increasing index set to I (i.e. $J \cup I \equiv \{1, 2, \cdots, n\}$ where J is also increasingly ordered).
2. $R(I,J) \in sym(n)$ rearranges $\{i_1, i_2, \cdots, i_k, j_1, j_2, \cdots, j_{n-k}\}$ in its increasing order.
3. For any $f \equiv \sum_{I \in \mathscr{T}_k} f^I dx^I$ we have

$$\star\, f := \sum_{I \in \mathscr{T}_k} f^I \star dx^I$$

An alternate way to define the *Hodge Star*: If f and g are $k-$forms, then $\star\, g$ is defined as the unique $(n-k)-$form such that

$$f \wedge \star g = \langle f, g \rangle dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n$$

where

$$\langle f, g \rangle := \sum_{I \in \mathscr{T}_k} f^I g^I$$

Lemma B.5.1. *Let $f \in \Omega(\Lambda^k)$. We then have*

$$\star \star f = (-1)^{k(n-k)} f$$

Proof. Since $\star$ is linear, it is enough to prove on basis elements.

$$\star dx^I = \mathrm{sign}(R(I,J)) dx^J$$

where J is the complement of I relative to $\{1, 2, \cdots, n\}$ and $R(I,J)$ is the permutation that rearranges $\{i_1, i_2, \cdots, i_k, j_1, j_2, \cdots, j_{n-k}\} \to \{1, 2, \cdots, n\}$.

$$\star \star dx^I = \mathrm{sign}(R(I,J)) \star dx^J = \mathrm{sign}(R(I,J))\, \mathrm{sign}(R(J,I)) dx^I$$

To compute the product of the signs we look at the map σ that takes (I,J), rearranges them to $\{1, 2, \cdots, n\}$ and then rearranges them to $\{J, I\}$. This amounts to moving the first k elements behind the last $n-k$ elements. This takes $k(n-k)$ transpositions, and hence the sign is $(-1)^{k(n-k)}$. Therefore,

$$\star\star\, dx^I = (-1)^{k(n-k)} dx^I$$

and the result follows for a general $k-$form from linearity.

■

Lemma B.5.2. *For $f, g \in \Omega(\Lambda^k)$*

$$\langle f, g\rangle = \langle \star f, \star g\rangle$$

Proof. Since $f \wedge \star g = \langle f, g\rangle dV$ (where dV is the volume form, in the Euclidean case, $dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n$), we have $\star f \wedge \star\star\, g = (-1)^{k(n-k)} \star f \wedge g = \langle \star f, \star g\rangle dV$. Recall that $f \wedge g = (-1)^{kl} g \wedge f$ for $f \in \Omega\Lambda^k$ and $g \in \Omega\Lambda^l$. Since $\star f \in \Lambda^{n-k}$ and $g \in \Lambda^k$, we have $\star f \wedge g = (-1)^{k(n-k)} g \wedge \star f$. This gives us

$$(-1)^{k(n-k)} \star f \wedge g = g \wedge \star f = \langle \star f, \star g\rangle dV$$

But since $g \wedge \star f := \langle g, f\rangle = \langle f, g\rangle$ (Real inner product), we have the result.

■

Appendix C

Manifolds